

Hyper Tuning Parameter

1.Decision Tree Regression

critrion	splitter	max_features	R score
<i>squared_error</i>	best	10	0.912
<i>squared_error</i>	best	0.5	-0.0125
<i>squared_error</i>	best	sqrt	0.653
<i>squared_error</i>	best	log2	0.354
<i>squared_error</i>	best	None	0.930
<i>squared_error</i>	random	2	0.310
<i>squared_error</i>	random	0.7	0.862
<i>squared_error</i>	random	sqrt	0.140
<i>squared_error</i>	random	log2	0.345
<i>squared_error</i>	random	None	0.914
<i>friedman_mse</i>	best	4	0.536

<i>friedman_mse</i>	best	0.4	0.207
<i>friedman_mse</i>	best	sqrt	0.618
<i>friedman_mse</i>	best	log2	0.882
<i>friedman_mse</i>	best	None	0.907
<i>friedman_mse</i>	random	7	0.674
<i>friedman_mse</i>	random	0.3	-0.0547
<i>friedman_mse</i>	random	sqrt	0.662
<i>friedman_mse</i>	random	log2	0.674
<i>friedman_mse</i>	random	None	0.904
<i>absolute_error</i>	best	8	0.932
<i>absolute_error</i>	best	0.2	0.0
<i>absolute_error</i>	best	sqrt	0.708
<i>absolute_error</i>	best	log2	0.568

<i>absolute_error</i>	best	None	0.961
<i>absolute_error</i>	random	3	0.850
<i>absolute_error</i>	random	0.8	0.792
<i>absolute_error</i>	random	sqrt	0.534
<i>absolute_error</i>	random	log2	0.507
<i>absolute_error</i>	random	None	0.918
<i>poisson</i>	best	25	0.927
<i>poisson</i>	best	0.678	0.770
<i>poisson</i>	best	sqrt	0.625
<i>poisson</i>	best	log2	0.764
<i>poisson</i>	best	None	0.9303
<i>poisson</i>	random	40	0.872
<i>poisson</i>	random	0.5	0.686

<i>poisson</i>	random	sqrt	0.697
<i>poisson</i>	random	log2	-1.55
<i>poisson</i>	random	None	0.916

For the following parameter combination :Criterion="Absolute error",splitter="best",max_features="None", R score=0.961, Decision tree Regression performs well and gives the maximum R score of 0.961

2.Random Forest Regression

critrion	n_estimators	max_features	R score
<i>squared_error</i>	10	sqrt	0.295
<i>squared_error</i>	100	sqrt	0.5622
<i>squared_error</i>	1000	sqrt	0.667
<i>squared_error</i>	10	log2	0.295
<i>squared_error</i>	100	log2	0.562
<i>squared_error</i>	1000	log2	0.667
<i>absolute_error</i>	10	sqrt	0.634
<i>absolute_error</i>	100	sqrt	0.641
<i>absolute_error</i>	1000	sqrt	0.701
<i>absolute_error</i>	10	log2	0.634

<i>absolute_error</i>	100	log2	0.641
<i>absolute_error</i>	1000	log2	0.701
<i>friedman_mse</i>	10	sqrt	0.295
<i>friedman_mse</i>	100	sqrt	0.562
<i>friedman_mse</i>	1000	sqrt	0.667
<i>friedman_mse</i>	10	log2	0.295
<i>friedman_mse</i>	100	log2	0.562
<i>friedman_mse</i>	1000	log2	0.667
<i>poisson</i>	10	sqrt	0.676
<i>poisson</i>	100	sqrt	0.579
<i>poisson</i>	1000	sqrt	0.621
<i>poisson</i>	10	log2	0.676
<i>poisson</i>	100	log2	0.579
<i>poisson</i>	1000	log2	0.621

For the following parameter combination: Criterion= “Absolute_error” ,
n_estimators=1000,max_features=”log2,sqrt”, Random forest Regressor performs well and gives a maximum Rscore of 0.701

3.SVM -Regression

Kernal	Gamma	C	R-score
Linear	scale	0.1	0.938
Linear	scale	10	0.565
Linear	scale	100	-7.92

Linear	auto	0.1	0.938
Linear	auto	10	0.565
Linear	auto	100	-7.92
poly	scale	0.1	-494778
poly	scale	10	-495
poly	scale	100	-2.665
rbf	scale	0.1	-38113675
rbf	scale	10	-380533
rbf	scale	100	-4265
rbf	auto	0.1	0
rbf	auto	10	0
rbf	auto	100	0
sigmoid	scale	0.1	-356709359602
sigmoid	scale	10	-35675245
sigmoid	scale	100	-357144
sigmoid	auto	0.1	0
sigmoid	auto	10	0
sigmoid	auto	100	0

*precomputed kernel works only with square matrix

For the parameter combination: Kernel="linear", Gamma="auto,scale",
C=0.1,SVR performs well with a Rscore of 0.938